GNSS Engineering Essentials

A TLC student's introduction to Satellite Navigation

Source: Kevin Stadnyk via Unsplash

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Authored by **Dario Blandi**
M.Sc. Engineering Student | University of Palermo

GNSS History

1973 - **NAVSTAR**: program launched by the US military for global navigation and missile guidance.

1983 - **Civilian Access**: system opened to public use for aviation safety.

1995 - **Full Capacity**: constellation of 24 satellites completed.

2000 - **High Precision**: intentional signal degradation removed; accuracy improved from $\sim 100m$ to $< 15m$.

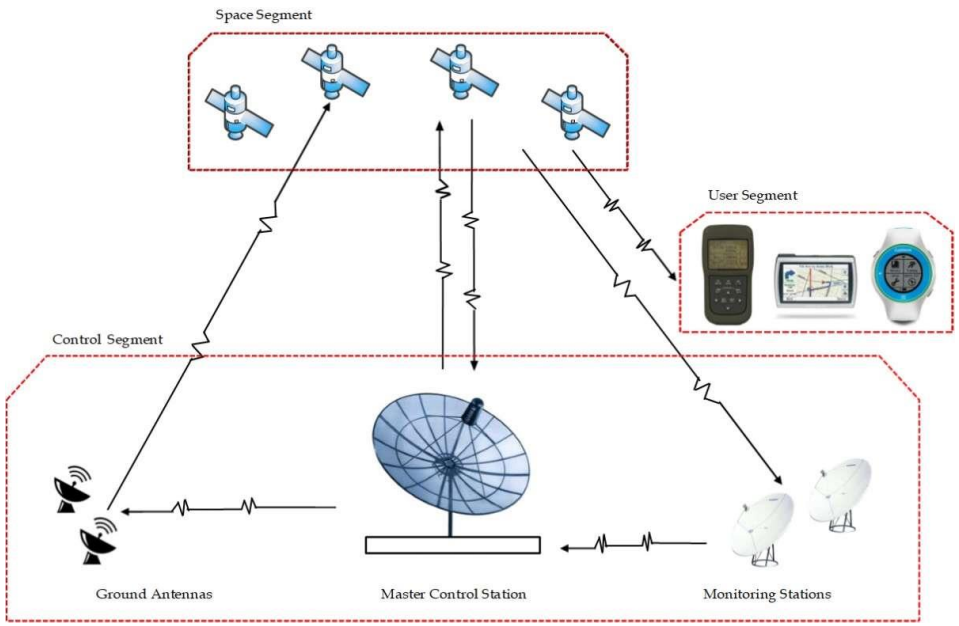
Source: U.S. Air Force

GNSS Architecture

Space Segment: constellation of 24+ satellites transmitting navigation signals and data.

Control Segment: Ground Stations monitoring satellite health, orbits, and clock synchronization.

User Segment: receivers (smartphones, cars, professional tools) that process signals to calculate position.



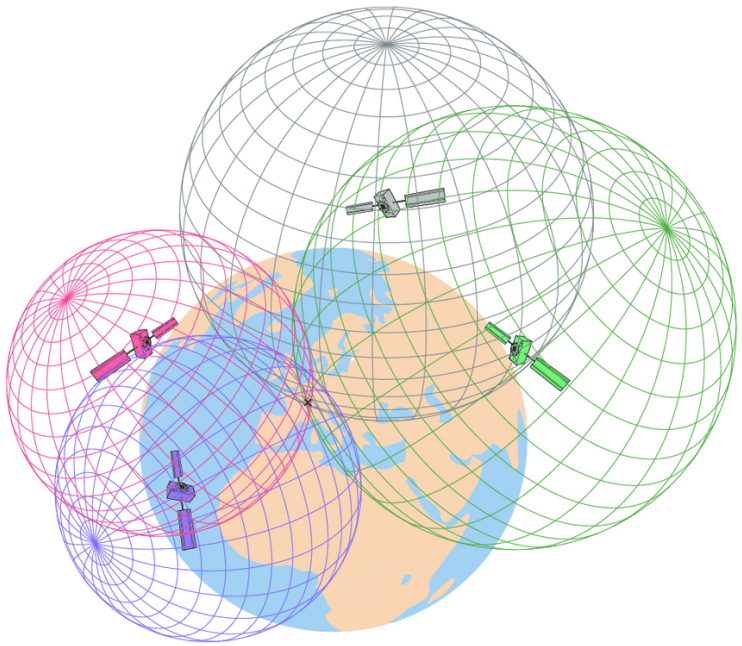
Source: IEEE Signal Processing Society

Positioning Fundamentals

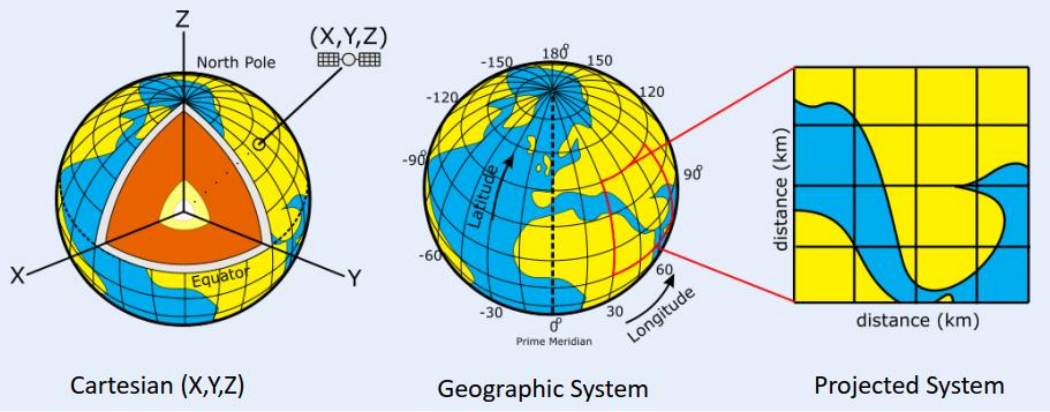
Trilateration: position is determined by measuring distances (ranges) from satellites at known locations.

Range Formula: $\boldsymbol{r} = \boldsymbol{c} \cdot \boldsymbol{\tau}$, where $\boldsymbol{c}$ is the speed of light and $\boldsymbol{\tau}$ is signal travel time.

- At least 4 satellites are required to solve for position $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}$ and the receiver's clock timing offset (Clock Bias).



Source: P.Y. Gilliéron (EPFL) via ResearchGate



Source: I. Lauer (Common Domain)

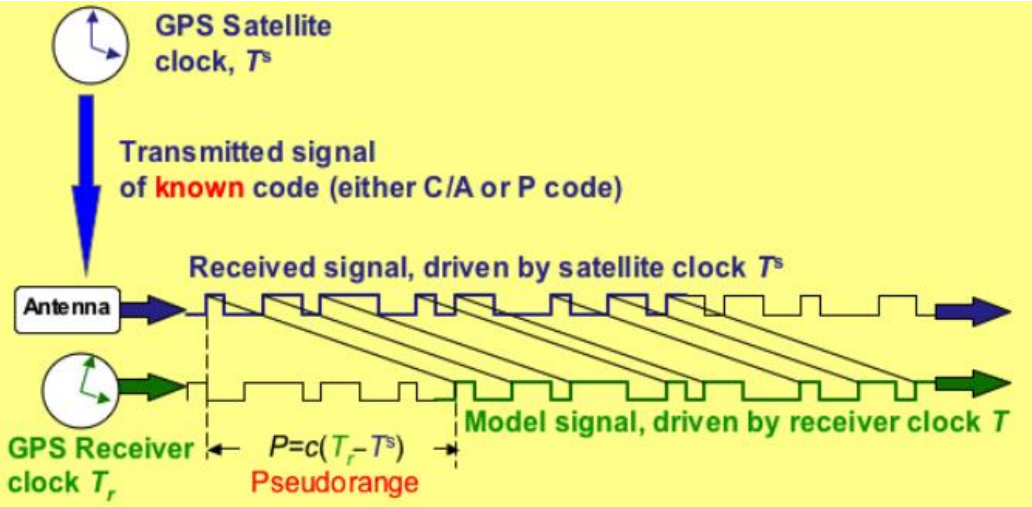
Measurements

Pseudorange: the apparent distance to a satellite, calculated by multiplying signal travel time by the speed of light.

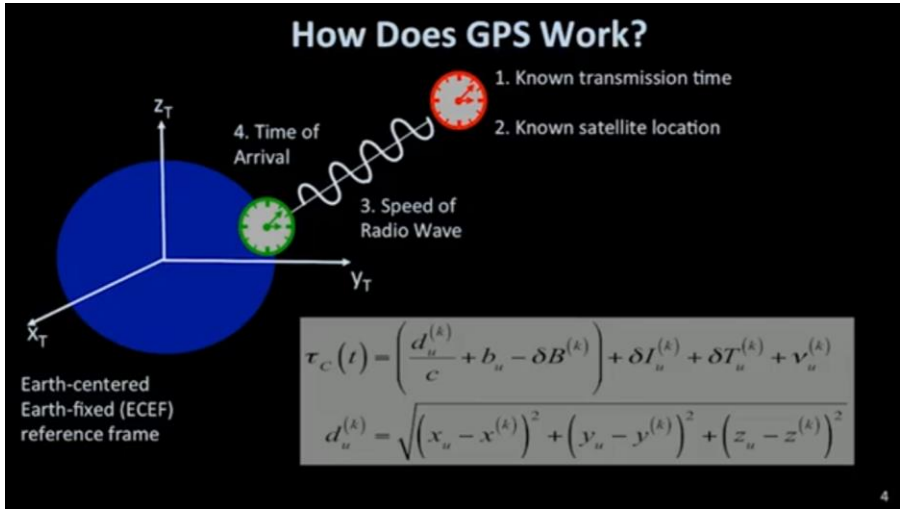
- Pseudorange is robust and unambiguous.

Carrier-Phase: for millimeter precision, the receiver can measure the carrier wave phase instead of the digital code delay.

- Carrier-Phase is extremely precise ($\sim \mathbf{0.01}\ \boldsymbol{m}$) but vulnerable to cycle slips.



Source: Cronin et al. (GETSI/NSF)



Source: Standford University

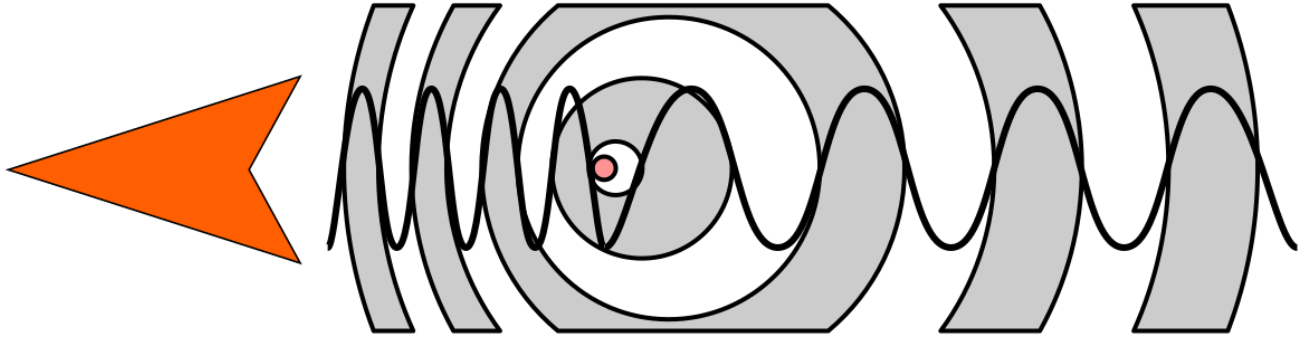
Signal Metrics

Doppler effect: frequency shift caused by relative motion between satellite and receiver.

- The receiver must constantly adjust to the shifting frequency to maintain a stable signal lock.

Carrier-to-Noise Density (C/N_0): a key measure of signal quality expressed in $dB\text{-}Hz$.

- C is the carrier received power $[W]$ and N_0 is the noise power density $[W/Hz]$.
- When expressed in logarithmic scale, the reference value for decibels, $1\ Hz$, is often denoted "$dB\text{-}Hz$".
- $\frac{C}{N_0} > 40\ dB\text{-}Hz$ usually indicates optimal signal quality.



Source: Wikimedia Commons

Space Key Features

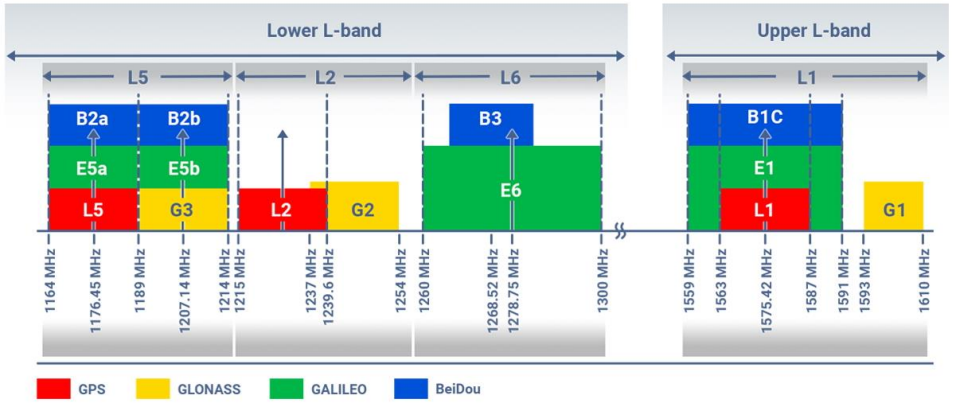
Global Constellations: GPS (USA), Galileo (EU), GLONASS (RU), BeiDou (CN).

***MEO* Altitudes**: $\sim 19000\ km$ to $> 23000\ km$.

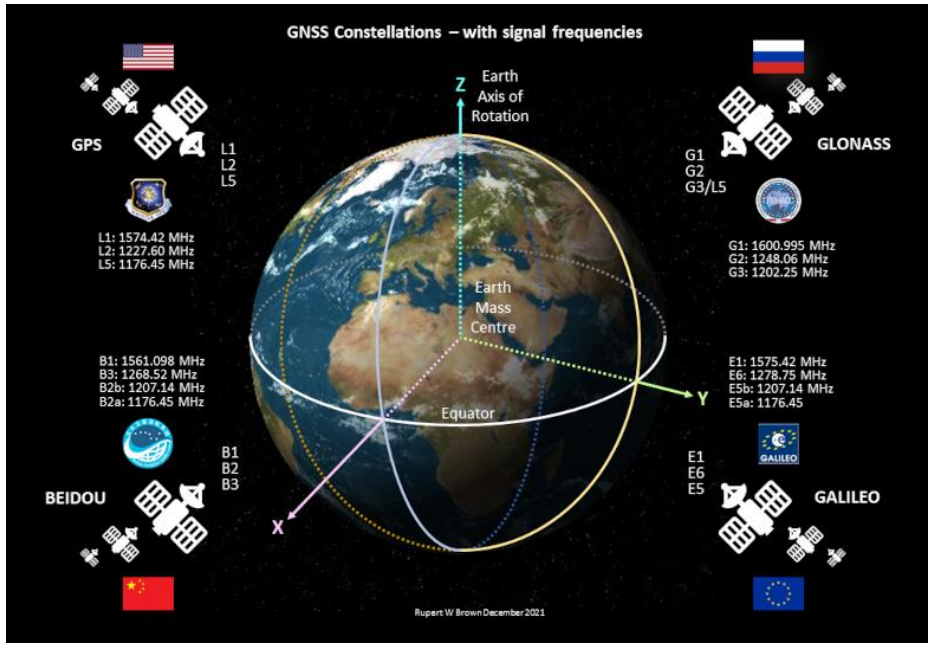
Atomic Clocks: high-stability Rubidium and Cesium standards.

Orbital Drifts: caused by solar radiation and gravity.

- Satellites under maintenance are marked as “out of service”.
- Receivers automatically ignore flagged satellites.



Source: Swift Navigation



Source: R. W. Brown

Global GNSS Constellations

GPS (USA)

- 24 nominal satellites in 6 orbital planes; inclination of $55°$.
- *DSSS* (Direct Sequence Spread Spectrum) *CDMA* (Code Division Multiple Access).
- Global visibility; at least 4 satellites are always visible from any point on Earth.

Galileo (EU)

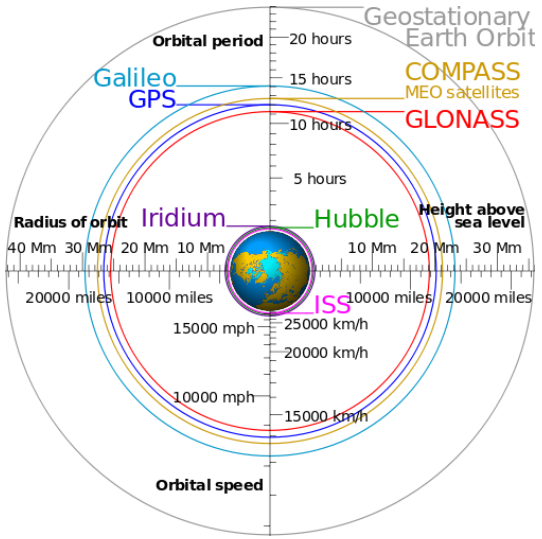
- 30 nominal satellites in 3 planes; inclination of $56°$.
- *DSSS CDMA*.
- The world's most accurate satellite navigation system.

GLONASS (Russia)

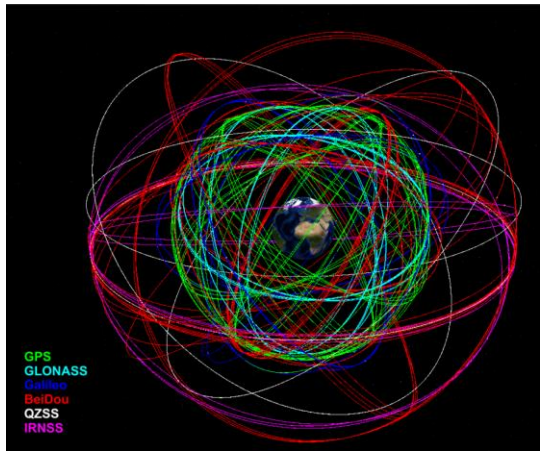
- 24 nominal satellites in 3 planes; inclination of $64.8°$.
- *DSSS FDMA* (Frequency Division Multiple Access) and *CDMA*.
- Optimized for high-latitude and polar regions.

BeiDou (China)

- 30 nominal satellites in 3 different orbits; inclination of $55°$ for *MEO* and *IGSO*, $0°$ for *GEO*.
- *DSSS CDMA*.
- High precision and enhanced services specifically for the China region.



Source: Wikimedia Commons



Source: GGOS / IAG

GPS Today's Satellites

Extended Longevity: design life has increased from 7.5 years to 15 years in the latest models.

Tri-band Capability: modern satellites broadcast across L1, L2, and L5 bands simultaneously.

Dual-purpose Payload: includes L-band antenna elements for signal transmission and S-band antennas for ground control.

Precision Propulsion: small rocket engines are used for precise orbit adjustments and "station-keeping".

Source: Lockheed Martin

GPS Control Segment

Global Layout: Master Control Station (*MCS*) and Alternate *MCS* are located in North America.

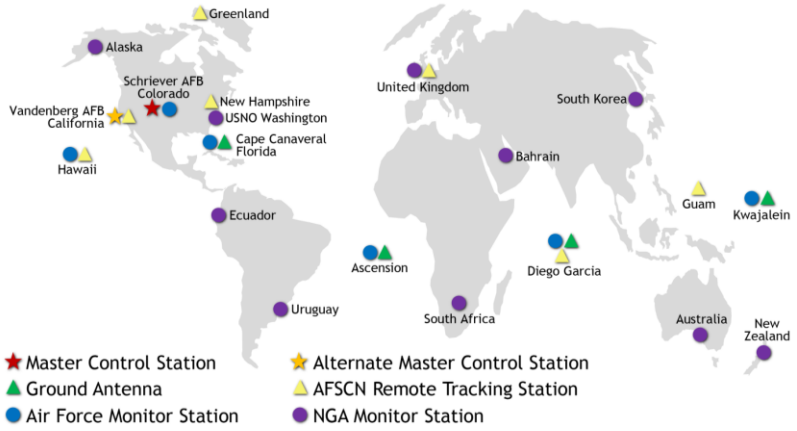
- There are 17 Monitoring Stations and 11 Ground antennas all over the world.

Trajectory & Clock Management: Monitor Stations relay tracking data to the *MCS*.

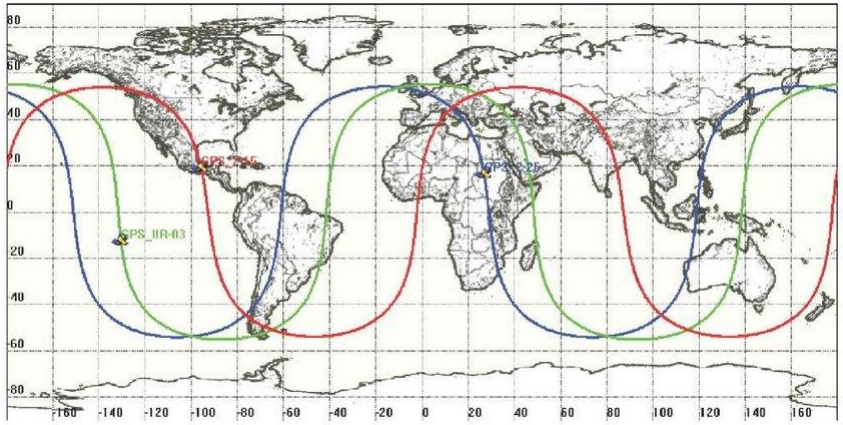
- *MCS* calculates Actual Trajectories to generate Predicted Trajectories.
- Navigation Messages are uploaded to satellites to update broadcast data and provide atomic clock corrections.

The Ground Track: projection of the satellite's orbit onto the Earth's surface.

- Due to Earth's rotation and the orbital inclination, the track forms a repeatable "wave" pattern.



Source: GPS.gov / U.S. Air Force



Source: Jahan et al. (2015) / ResearchGate

GPS Receiver Architecture

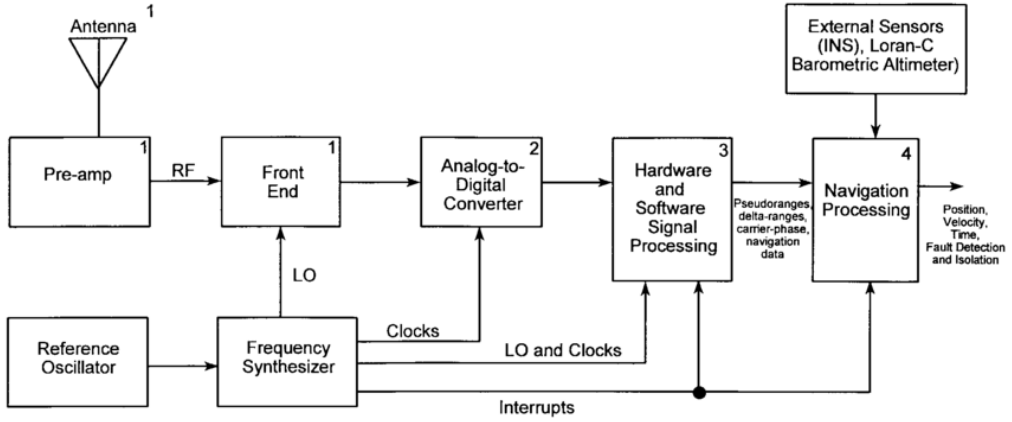
Antenna and Preamplifier: uses Circular Polarization and a Low Noise Amplifier to minimize the noise.

Front-End: down-converts High Frequency signals to an Intermediate Frequency (*IF*).

Internal Clock: provides the local timing reference for signal Correlation.

Tracking Loops: *DLL* (Delay Lock Loop) and *PLL* (Phase Lock Loop) to synchronize with satellite code and carrier phase.

Navigation Processing (*PVT*): final conversion of raw measurements into Position, Velocity, and Time.



Source: Braasch & van Dierendonck (1999)

GPS Legacy Signals

Source: GPS World

SVN	PRN	CLOCK	LAUNCHED	USABLE	PLANE/SLOT	NOTES
TYPE: Block IIA						
34	18	Rb	10-26-93	3-20-18	D6	A
TYPE: Block IIR						
43	13	Rb	7-23-97	1-31-98	F6	
46	11	Rb	10-7-99	1-3-00	D2-F	
51	20	Rb	5-11-00	6-1-00	E4	B
44	28	Rb	7-16-00	8-17-00	B3	
41	14	Rb	11-10-00	12-10-00	F2-F	
54	18		1-30-01	2-15-01		C
56	16	Rb	1-29-03	2-18-03	B1-A	
45	21	Rb	3-31-03	4-12-03	D3	
47	22	Rb	12-21-03	1-12-04	E6	
59	19	Rb	3-20-04	4-5-04	C5	
60	23	Rb	6-23-04	7-9-04	F4	
61	02	Rb	11-6-04	11-22-04	D1	
TYPE: Block IIR-M						
53	17	Rb	9-26-05	12-16-05	C4	
52	31	Rb	9-25-06	10-12-06	A2	
58	12	Rb	11-17-06	12-13-06	B4	
55	15	Rb	10-17-07	10-31-07	F2-A	
57	29	Rb	12-20-07	1-2-08	C1	
48	07	Rb	3-15-08	3-24-08	A4	
50	05	Rb	8-17-09	8-27-09	E3	
TYPE: Block IIF						
62	25	Rb	5-28-10	8-27-10	B2	
63	01	Rb	7-16-11	10-14-11	D2-A	
65	24	Cs	10-4-12	11-14-12	A1	
66	27	Rb	5-15-13	6-21-13	C2	
64	30	Rb	2-21-14	5-30-14	A3	
67	06	Rb	5-17-14	6-10-14	D4	
68	09	Rb	8-2-14	9-17-14	F3	
69	03	Rb	10-29-14	12-12-14	E1	
71	26	Rb	3-25-15	4-20-15	B1-F	
72	08	Cs	7-15-15	8-12-15	C3	
73	10	Rb	10-31-15	12-9-15	E2	
70	32	Rb	2-5-16	3-9-16	F1	

L1 Signal: combines both the *P*-code and the *C/A*-code.

$$s_{L1}(t) = A_{L1}Y(t)N(t)\cos(w_{L_1}t) + A_{C/A}CA(t)N(t)\sin(w_{L_1}t)$$

L2 Signal: in its legacy form, it contains only the P-code.

$$s_{L2}(t) = A_{L2}Y(t)N(t)\,cos(w_{L_2}t)$$

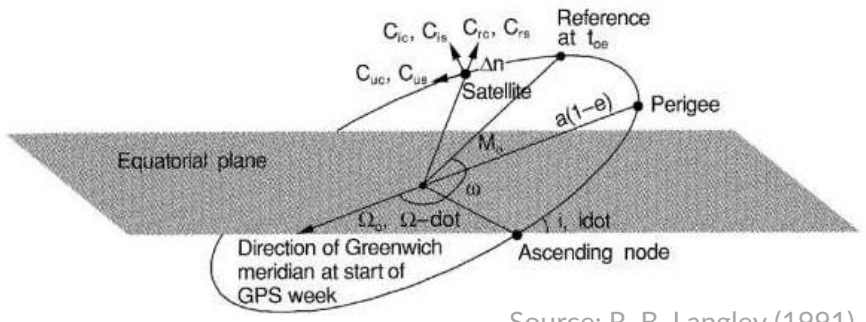
Navigation Message $N(t)$: a slow $\mathbf{50}\ \boldsymbol{bps}$ data stream providing system time, satellite status, Ephemeris and Almanac.

Ephemeris: highly precise orbital and clock data for a specific satellite.

- Required for accurate positioning and updated every few hours.

Almanac: low precise orbital data for the entire constellation.

- Used by the receiver to quickly locate visible satellites during startup.



Source: R. B. Langley (1991)

GPS PRN Codes

Pseudo-Random Noise (*PRN*) Code: a sequence of fast "Chips" representing a single bit, used to identify every satellite.

- Each Code is unique and orthogonal for each satellite.
- The *C/A* (Coarse/Acquisition) *PRN* civil Code operates at a $\mathbf{1.023}\ \boldsymbol{MHz}$ chipping rate; repeats every $\mathbf{1}\ \boldsymbol{ms}$ for fast acquisition.
- The *P* (Precise) PRN military Code operates at a $\mathbf{10.23}\ \boldsymbol{MHz}$ chipping rate; repeats every week and can be encrypted as $\boldsymbol{P}(\boldsymbol{Y})$.

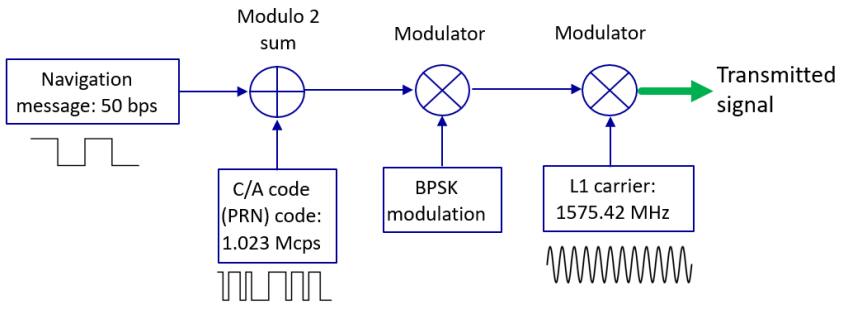
Direct Sequence Spread Spectrum (*DSSS*): multiplies (*XOR*) the slow Navigation Message $\boldsymbol{N}(\boldsymbol{t})$ by the fast *PRN* Chips.

- This technique spreads the signal bandwidth to make it more robust to interference.

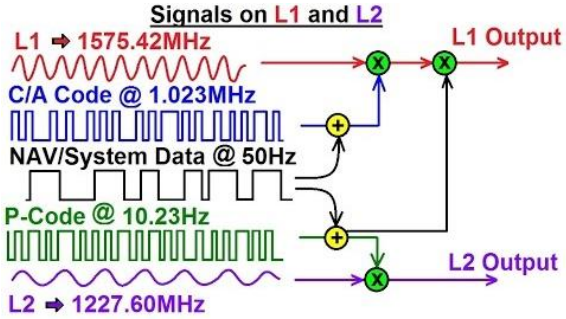
Code Division Multiple Access (*CDMA*): multiple satellites can share the same frequency thanks to *DSSS*.

- The receiver can filter the desired *PRN* Code via Correlation.

Biphase Shift Key (*BPSK*) Modulation: Chips modulate the carrier wave phase ($+\mathbf{1} = \mathbf{0}°$ shift, $-\mathbf{1} = \mathbf{180}°$ shift).



Source: Wasy Research (2021)



Source: M. van Biezen (iLectureOnline)

GPS Carrier Frequencies

Fundamental Frequency: $\boldsymbol{f_0 = 10.23\ MHz}$.

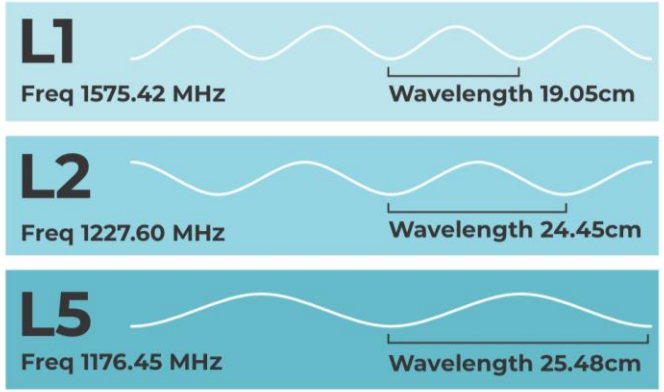
- All *GPS* carrier frequencies and chipping rates are derived as multiples of $\boldsymbol{f_0}$.

Legacy Signals (Original Bands)

- L1 ($\boldsymbol{f_{L_1} = 154\ f_0 = 1575.42\ MHz}$) used for the original *C/A* and $\boldsymbol{P(Y)}$ codes.
- L2 ($\boldsymbol{f_{L_2} = 120\ f_0 = 1227.60\ \mathrm{MHz}}$) originally used for military $\boldsymbol{P(Y)}$ code; now used for dual-frequency ionospheric corrections.

Modernized Signals (New Generations)

- L5 ($\boldsymbol{f_{L_5} = 115\ f_0 = 1176.45\ \mathrm{MHz}}$) used for "Safety-of-Life" use cases with higher power and more interference resistance.
- L1C / L2C used for better tracking in "urban canyons" and compatibility with Galileo.
- M-Code used on L1 / L2 for maximum security and anti-jamming.



Source: SparkFun Electronics

Propagation and Errors

Atmospheric Delays

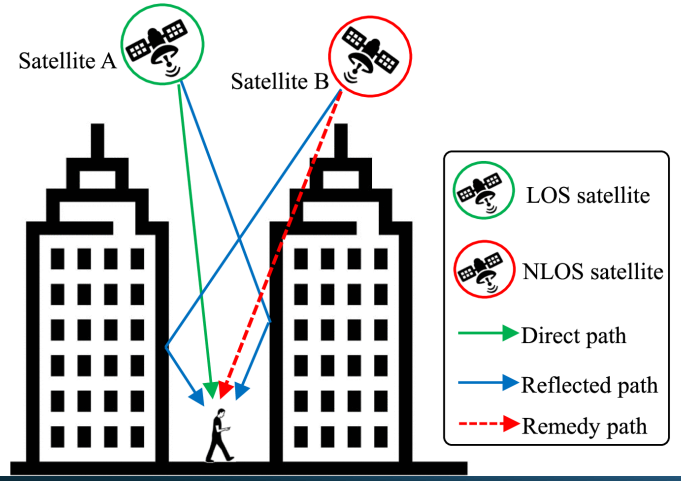
- Charged particles in the Ionosphere ($50 - 1000\ km$) refract the signal, altering its speed and transmission time.
- Local weather (humidity, temperature, pressure) in Troposphere ($0 - 50\ km$) causes small, variable delays.

Multipath Error: signals bounce off buildings or walls before reaching the antenna.

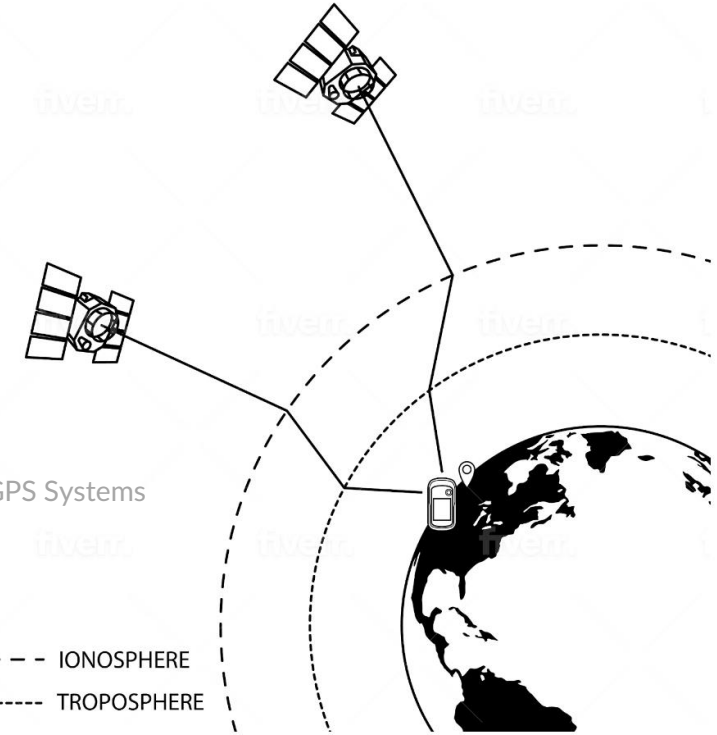
- The reflected path is longer, leading to delayed arrival and incorrect distance calculations.

Line of Sight (*LOS*): *GNSS* requires a direct, straight path between satellite and receiver.

Urban Canyons: skyscrapers, trees, and bridges can block or degrade signals entirely.



Source: IEEE Computer Society



Source: Global GPS Systems

Geometric Accuracy

Dilution of Precision (*DOP*): numerical factor representing how satellite positions in the sky affect positioning accuracy.

- Low *DOP* indicates that satellites are widely spaced; small area of uncertainty.
- High *DOP* indicates that satellites are clustered together; large area of uncertainty.

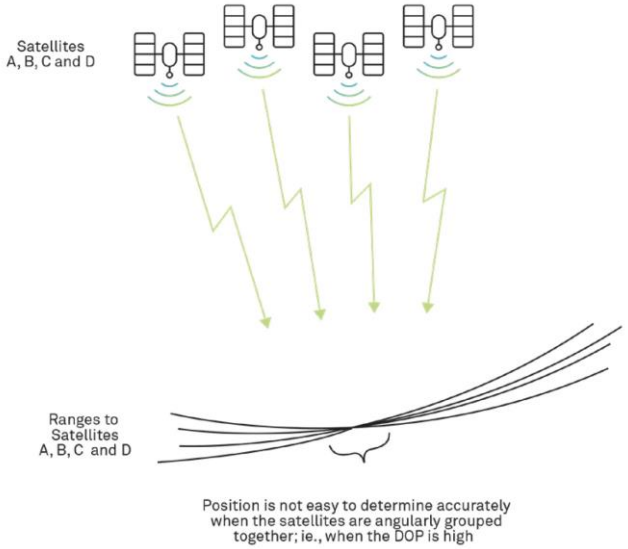
***GDOP* (Geometric)**: total uncertainty (3D position + Time).

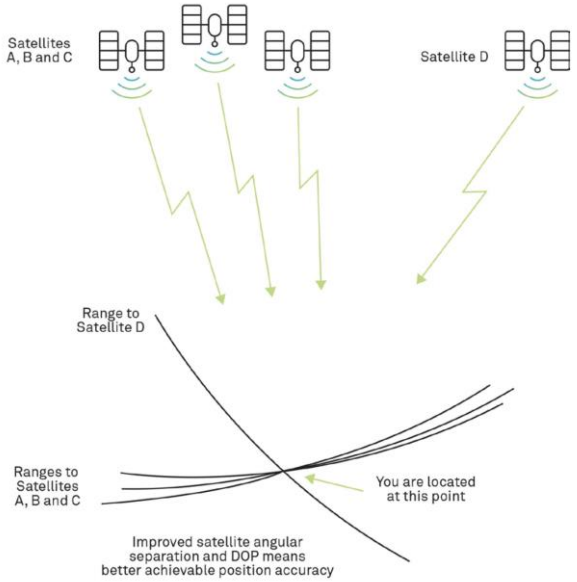
***PDOP* (Positional)**: 3D uncertainty (Latitude, Longitude, Height).

- $\boldsymbol{PDOP < 4}$ usually indicates excellent accuracy.

HDOP / VDOP: Horizontal and Vertical uncertainty.

***TDOP* (Time)**: receiver clock offset uncertainty.





Source: NovAtel (Hexagon)

Error Sources

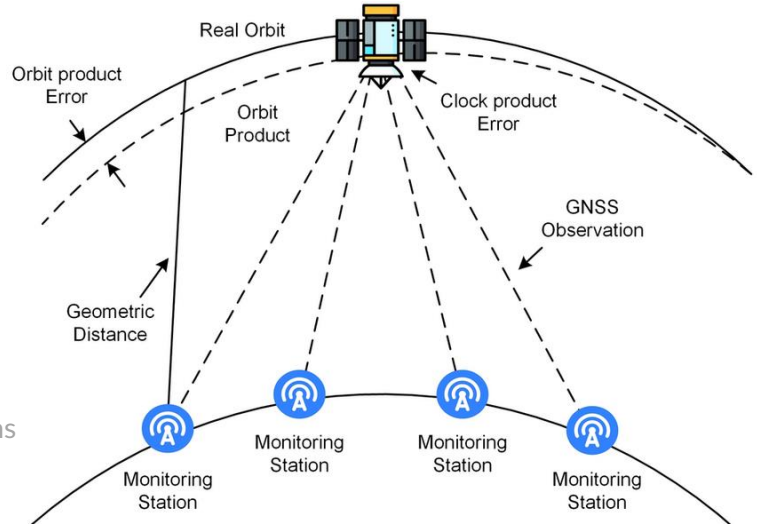
Propagation speed: radio waves travel at the speed of light; a $\mathbf{1}\ \boldsymbol{ns}$ timing error translates to a $\mathbf{30}\ \boldsymbol{cm}$ position error.

Atomic Clock Drift: despite extreme stability, slight drifts contribute $\pm\mathbf{2}\ \boldsymbol{m}$ of error.

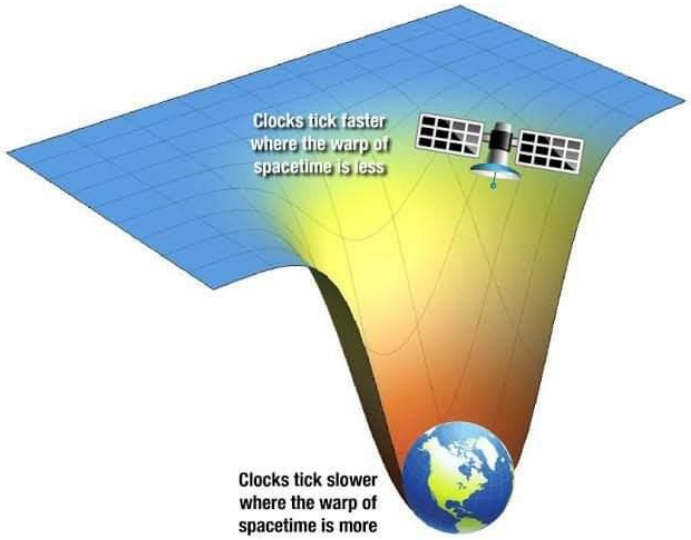
General Relativity: satellites are in a weaker gravitational field, causing clocks to run faster ($+\mathbf{45}\ \boldsymbol{\mu s/day}$).

Special Relativity: high orbital speeds ($\sim\mathbf{3}\ \boldsymbol{km/s}$) cause clocks to run slower ($-\mathbf{7}\ \boldsymbol{\mu s/day}$).

Orbit Errors: discrepancies between actual and predicted satellite positions ($\pm\mathbf{2.5}\ \boldsymbol{m}$).



Source: Huang et al. (2024) / GPS Solutions



Source: Write Science

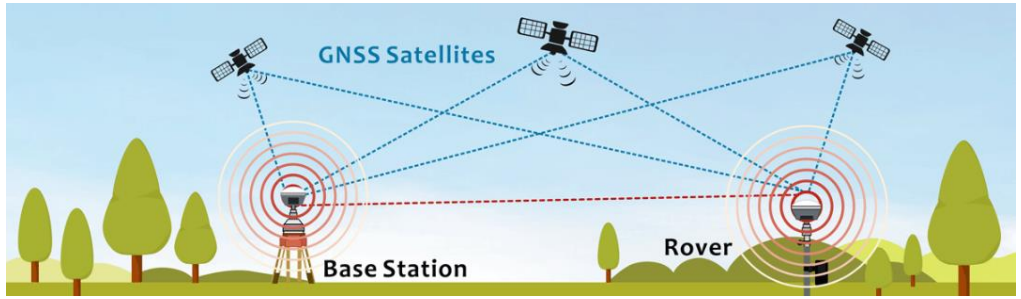
High-Precision Techniques

Differential Correction (*DGNSS*): a fixed Base Station calculates signal errors and transmits instant corrections to a Rover.
- Corrections that are made with *DGNSS* achieve sub-meter accuracy.

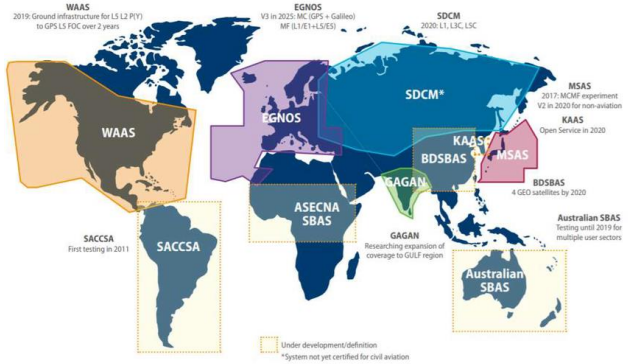
Satellite Based Augmentation System (*SBAS*): geostationary satellites broadcast wide-area corrections.
- SBAS is primarily used for aviation safety.

Real-Time Kinematic (*RTK*): a type of *DGNSS* that measures the Carrier Phase of the signal instead of just the *PRN* code.
- Corrections that are made with *RTK* achieve centimeter-level precision yet require a stable data link.
- Network *RTK* uses differential corrections sent to Rovers through Internet.

Precise Point Positioning (*PPP*): a single independent receiver uses a worldwide network to model precise corrections.
- *PPP* corrections provide global decimeter-to-centimeter accuracy.



Source: Unictron Technologies Corp.



Source: Siddiqui et al. (2022) / ResearchGate

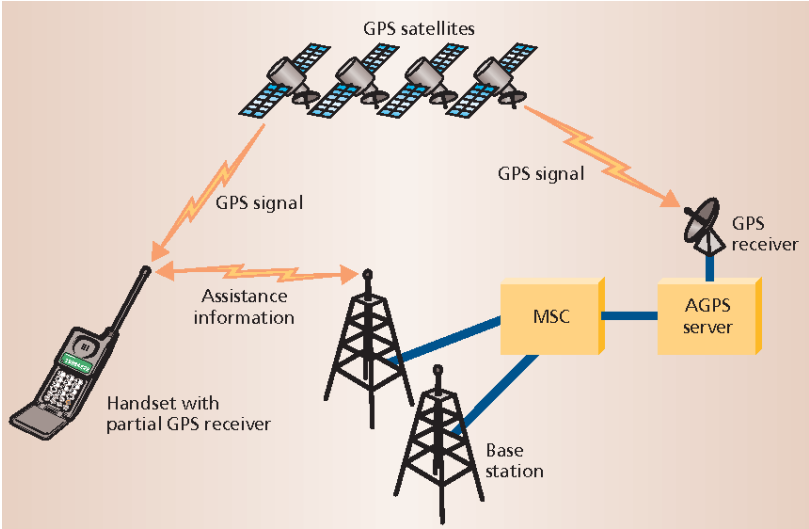
Hybrid Systems

Assisted GNSS (*A-GNSS*): terrestrial cellular networks provide the navigation message via a server.

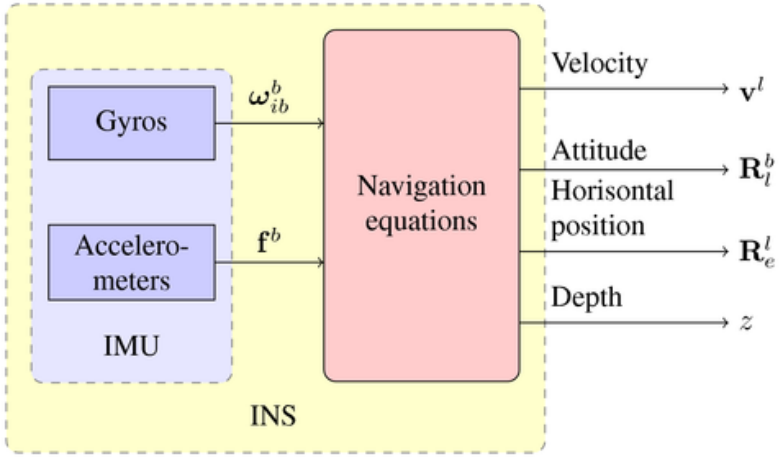
- This enables significantly faster signal acquisition and tracking even at very low signal-to-noise ratios.

Inertial Navigation System (*INS*): uses accelerometers and gyroscopes to compute a relative position.

- *GNSS* + *INS* is used in urban canyons or tunnels where the Line of Sight is frequently blocked.



Source: Djuknic & Richton (2001) / IEEE



Source: Rajput et al. (2019) / INCAS Bulletin

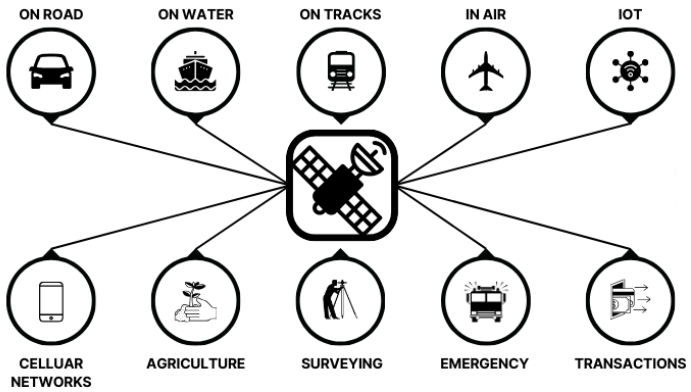
Applications and Equipment

Main Applications

- Centimeter-level accuracy for Surveying, Precision Agriculture, and autonomous Machine Control.
- Mission-critical reliability for Aviation, Marine, and Defense operations.
- Location services for Smartphones, Road Navigation, and high-precision Timing for financial networks.

Hardware Selection Criteria

- Accuracy requirements, Signal Availability, and fast acquisition Time To First Fix (*TTFF*).
- Resistance to environmental factors like water/dust, shock, and vibration.
- Low Power Consumption and portability, essential for UAVs (Drones) and wearables.
- Support for Multi-Constellation and Multi-Frequency chipsets.



Source: Ali et al. (2024) / ResearchGate

Source: Amazon

The Future of SatCom

Free-Space Optical Communication (*FSO*)

- The use of precise lasers can achieve Terabit-per-second speeds and quantum-level security.
- E.g. HydRON project is set to to skyrocket the speed and security of space-based communications while reducing costs.

Lunar Communications and Navigation Services (*LCNS*)

- Stable Moon links will enable precision landing, surface mobility, and high-speed data for human habitats and robotic missions.
- E.g. Moonlight Programme aims to enstablish a permanent constellation around the Moon for the future lunar economy.

5G Non-Terrestrial Networks (*NTN*)

- Integrating satellites into the 5G ecosystem extends cellular coverage to remote and maritime zones.
- E.g. Celeste is a "Direct-to-Device" solution allowing standard smartphones to connect directly to satellites.

Source: European Space Agency